POWER BI PROJECT

E-Commerce Sales & Customer Behaviour Analysis

An interactive dashboard to explore sales performance, customer patterns, product categories and shopping devices.

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Tool: Microsoft Power BI Desktop

1. Project Overview

Purpose, scope and business questions

PROJECT OBJECTIVE

To turn e-commerce transaction data into clear visual insights that support better business decisions.

KEY QUESTIONS

- How do sales change over time?
- Which product categories generate the most sales?
- Which cities contribute most to sales?
- How are orders distributed across devices?

Outcome: a single-page dashboard with KPIs, charts and interactive filters.

2. Dataset & Methodology

A practical overview of the data and workflow

DATASET

- 5,000 transaction records
- 18 fields / columns
- Period: Jan 2023–Mar 2024
- Currency: Turkish Lira (TRY)

IMPORTANT FIELDS

- Order and customer IDs
- Date, city, gender, age
- Product, price, quantity, discount
- Device, payment, rating, delivery

WORKFLOW

- Imported CSV into Power BI
- Removed duplicate query
- Removed blank rows
- Created measures and visuals

Note: Dataset figures and descriptions should be checked against the final imported file before submission.

3. KPI Snapshot

Key measures shown on the dashboard

TOTAL SALES	TOTAL ORDERS	TOTAL CUSTOMERS
4.92M	5K	5K

AVERAGE ORDER VALUE	RETURNING CUSTOMER	AVERAGE RATING
983.11	59.80%	3.90

KPIs summarise overall performance. Values are taken from the current dashboard and may change when filters are applied.

4. Dashboard Visuals & Interactivity

How the dashboard helps users explore performance

Monthly Sales Trend

Tracks changes in sales across the timeline.

Sales by Product Category

Compares category-level sales; Electronics is the largest bar in the current view.

Sales by City

Highlights differences between city sales; Istanbul is the largest bar in the current view.

Orders by Device Type

Compares Mobile, Desktop and Tablet order counts; Mobile is highest in the current view.

Filters: City • Gender • Device Type • Product Category

5. Insights & Recommendations

Use the visuals to guide further investigation

OBSERVATIONS FROM THE CURRENT VIEW

- Electronics leads category sales in the current view.
- Istanbul has the highest sales bar among displayed cities.
- Mobile accounts for the most orders across device types.
- The dashboard reports a 3.90 average customer rating.

RECOMMENDATIONS

- Review stock and promotions for high-performing categories.
- Study city-level demand before planning local campaigns.
- Make the mobile shopping journey fast and easy to use.
- Compare ratings and delivery times to identify service gaps.

Conclusion: Power BI makes key business patterns easier to see and discuss. Findings describe associations, not proof of cause and effect.